Customer Churn Analysis — R Code Documentation

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Script: `churn_analysis.R`

Environment: R 4.6.1, RStudio

Purpose

The script analyses a subscriptions dataset to:

1. Describe statistical differences between subscribers who churned and those who stayed.
2. Identify the three strongest factors associated with churn.
3. Estimate the effect of each factor with 95% confidence intervals.
4. Evaluate the model with metrics suited to imbalanced data (churn rate 9.3%).

The results are associational, not causal: the data are cross-sectional and observational.

Data

Item	Value
File	`subscriptions-churn.csv` (UTF-8, Arabic headers and category values)
Rows	2,400 subscribers, one row per subscriber
Columns	12
Outcome	`Churned` (1 = cancelled, 0 = retained); 223 churned (9.3%)
Quality	No missing values, no duplicate subscriber IDs

Variable (English name)	Original header	Role
`Churned`	ألغى الاشتراك	Outcome
`Plan`	الباقة	Factor (reference: أساشي / Basic)
`BillingCycle`	دورة الفوترة	Factor (reference: شهري / Monthly)
`PaymentMethod`	وسيلة الدفع	Factor (reference: بطاقة / Card)
`AttendedOnboarding`	حضر التهيئة	Factor (1/0)
`DiscountRate`	نسبة الخصم	Factor (modelled per 5 percentage points)
`SupportTickets`	تذاكر الدعم	Factor (spline)
`AvgMonthlySessions`	متوسط الجلسات الشهرية	Factor (spline)
`MonthlyPrice`	السعر الشهري	Excluded: identical to Plan
`TenureMonths`	أشهر البقاء	Descriptive only
`SubscriptionDate`	تاريخ الاشتراك	Descriptive only
`SubscriberID`	معرف المشترك	Identifier only

Setup

Install the packages once:

```
install.packages(c("tidyverse", "broom", "binom", "car", "pROC",
                   "PRROC", "marginaleffects", "effectsize"))
```

`splines` ships with base R. Before running, update the file path in Step 1 to where the CSV is saved. Then run the script from top to bottom; each step depends on objects created in the steps before it.

Script structure

Step	What it does	Main methods	Key objects
1. Data preparation	Reads the CSV, renames columns, converts flags to 1/0 and discount to 0–0.25, sets reference categories, checks counts against Power BI	`read_csv`, `mutate`, `factor`	`df`
2. Descriptive statistics	Compares churned vs retained on every variable	Mann-Whitney U + rank-biserial; chi-square + bias-corrected Cramér's V	`num_summary`, `num_tests`, `cat_tests`
3. Churn rates	Churn rate per category with Wilson 95% CI, and difference from the reference level in percentage points	`binom.confint` (Wilson), `prop.test`	`results3`
4. Logistic regression	Compares linear vs spline model, ranks factors, estimates odds ratios and average marginal effects, checks collinearity, plots adjusted curves	`glm`, `ns`, `anova`, `drop1`, `tidy`, `avg_comparisons`, `vif`, `plot_predictions`	`m_linear`, `m_spline`, `ame_cat`, `ame_num`
5. Model evaluation	Stratified 5-fold cross-validation; discrimination, calibration and threshold metrics	`roc`, `ci.auc`, `pr.curve`, Brier score, deciles	`df$pred_cv`, `roc_obj`, `pr`
6. Mechanism test	Tests whether onboarding is linked to support tickets	Mann-Whitney U	—

Analytical decisions

Decision	Reason
No resampling (SMOTE, oversampling)	The analysis explains factors; resampling distorts real churn rates and regression estimates
Wilson intervals for churn rates	More accurate than the normal approximation when rates are near zero
Effect sizes reported beside p-values	Significance alone does not show how large a difference is
Bonferroni threshold noted (0.05 / 8 = 0.00625)	Eight tests in Step 2 raise the chance of a false positive
Practical importance judged on percentage-point differences	Cramér's V stays small under imbalance even when differences matter
`MonthlyPrice` excluded	One fixed price per plan, so it duplicates `Plan`

Decision	Reason
`TenureMonths` excluded from the model	Inconsistent with `SubscriptionDate`, and measures time-to-churn for churners but time-so-far for others
Splines for sessions and tickets	Step 3 showed a U-shape for sessions; the spline model fits significantly better (LRT = 33.6, 4 df, p < 0.001; AIC 1405.8 vs 1431.5)
Factor ranking by AIC increase on deletion	Fair across terms with 1 vs 3 degrees of freedom
Pre-specified contrasts (sessions 30 vs 10, tickets 8 vs 2)	The effect of a curved variable depends on the points compared
55 vs 28 sessions reported as exploratory	The comparison point was chosen after seeing the curve
Stratified 5-fold cross-validation, `set.seed(2026)`	Honest out-of-sample evaluation; reproducible folds
Accuracy reported only next to recall, precision and PR-AUC	A model predicting "no churn" for everyone reaches 90.7% accuracy

Key results

Three strongest factors (adjusted model, average marginal effect on churn probability):

Rank	Factor	Contrast	Effect (pp)	95% CI
1	Monthly sessions	30 vs 10 sessions	−7.0	−9.3 to −4.6
2	Support tickets	8 vs 2 tickets	+7.9	+4.6 to +11.2
3	Onboarding attendance	Attended vs not	−4.1	−6.4 to −1.8

Plan (Advanced/Business vs Basic: −4.0 and −3.5 pp) and annual billing (−3.2 pp) also show smaller effects. Payment method and discount show no association.

Model evaluation (cross-validated):

Metric	Value	Baseline
ROC-AUC	0.682 (95% CI 0.646–0.719)	0.5
PR-AUC	0.180	0.093
Brier score	0.0812	0.0843
Top 20% risk: recall / precision	40.4% / 18.8%	— / 9.3%

The model ranks subscribers by risk (about twice the churn rate of random targeting in the top 20%) but is not precise enough for individual prediction.

Limitations

- Associational results only; causal claims need an experiment.
- The source does not define sessions, onboarding or the subscription date.
- `TenureMonths` is inconsistent with `SubscriptionDate` (67% of tenures exceed the time elapsed since subscription), so timing of churn cannot be analysed.
- The near-zero correlations between factors (VIF ≈ 1.00) suggest the data may be synthetic.

Appendix: Full R Script (churn_analysis.R)

The complete script, run top to bottom in RStudio. Comment lines are shown in green.

```
# ============================================================================
# Customer Churn Analysis — Subscriptions Dataset
# Author : Leena A. Elsheikh
# R      : 4.6.1
# Purpose: Describe differences between churned and retained subscribers,
#          identify the three strongest factors associated with churn,
#          estimate each factor's effect with confidence intervals, and
#          evaluate the model with metrics suited to imbalanced data.
#
# Data   : subscriptions-churn.csv — 2,400 subscribers, one row each,
#          12 columns (Arabic headers and Arabic category values).
#
# One-time setup (run once in the Console, not every time):
# install.packages(c("tidyverse", "broom", "binom", "car", "pROC",
#                    "PRROC", "marginaleffects", "effectsize"))
# ============================================================================


# ============================================================================
# STEP 1 — Data preparation
# Mirrors the cleaning done in Power Query, so results match Power BI.
# ============================================================================
library(tidyverse)

# 1.1 Read the file (UTF-8 so Arabic text is read correctly)
df <- read_csv("C:/Users/lina0/Downloads/subscriptions-churn.csv",
               locale = locale(encoding = "UTF-8"))

# 1.2 English column names (same order as the columns in the file)
names(df) <- c("SubscriberID", "SubscriptionDate", "Plan", "MonthlyPrice",
               "BillingCycle", "TenureMonths", "PaymentMethod", "SupportTickets",
               "AvgMonthlySessions", "DiscountRate", "AttendedOnboarding", "Churned")

# 1.3 Transformations
#   - DiscountRate: 0–25 -> 0–0.25
#   - Yes/No (نعم/لا) flags -> 1/0, so the mean of each column is a rate
#   - Factor level order sets the reference category in the regression:
#     Plan = Basic (أساسي), BillingCycle = Monthly (شهري), PaymentMethod = Card (بطاقة)
df <- df %>%
  mutate(
    DiscountRate       = DiscountRate / 100,
    AttendedOnboarding = as.integer(AttendedOnboarding == "نعم"),
    Churned            = as.integer(Churned == "نعم"),
    Plan          = factor(Plan, levels = c("أساسي", "متقدم", "أعمال")),
    BillingCycle  = factor(BillingCycle, levels = c("شهري", "سنوي")),
    PaymentMethod = factor(PaymentMethod, levels = c("بطاقة", "محفظة", "تحويل"))
  )

# 1.4 Checks (expected: 2400 rows, 223 churned = 0.0929, 1314 attended onboarding)
glimpse(df)
nrow(df)
sum(df$Churned); mean(df$Churned)
sum(df$AttendedOnboarding)
table(df$Plan); table(df$BillingCycle); table(df$PaymentMethod)
sort(unique(df$DiscountRate))
```

```r
class(df$SubscriptionDate)


# ============================================================================
# STEP 2 — Descriptive statistics: churned vs retained
# Numeric: Mann-Whitney U + rank-biserial effect size.
# Categorical: chi-square + bias-corrected Cramér's V.
# TenureMonths is descriptive only (inconsistent with SubscriptionDate and
# related to the outcome itself), so it is not used in the model.
# ============================================================================
library(effectsize)

# Group label for display
df <- df %>%
  mutate(Group = factor(Churned, levels = c(0, 1), labels = c("بقي", "ألغى")))

# 2.1 Numeric variables: summary by group
num_vars <- c("AvgMonthlySessions", "SupportTickets", "DiscountRate", "TenureMonths")

num_summary <- df %>%
  group_by(Group) %>%
  summarise(across(all_of(num_vars),
                   list(mean = mean, median = median, sd = sd),
                   .names = "{.col}_{.fn}")) %>%
  pivot_longer(-Group, names_to = c("Variable", "Stat"), names_sep = "_") %>%
  pivot_wider(names_from = c(Group, Stat), values_from = value)

print(num_summary, width = Inf)

# 2.2 Numeric variables: test and effect size
# Positive r_rb = retained group higher; negative = churned group higher
num_tests <- map_dfr(num_vars, function(v) {
  f  <- reformulate("Group", response = v)
  w  <- wilcox.test(f, data = df, exact = FALSE)
  rb <- rank_biserial(f, data = df)
  tibble(Variable = v,
         p_value  = w$p.value,
         r_rb     = rb[[1]],
         CI_low   = rb$CI_low,
         CI_high  = rb$CI_high)
})

print(num_tests, width = Inf)

# 2.3 Categorical variables: distribution within each group (column %)
cat_vars <- c("Plan", "BillingCycle", "PaymentMethod", "AttendedOnboarding")

for (v in cat_vars) {
  cat("\n====", v, "====\n")
  tab <- table(df[[v]], df$Group)
  print(round(100 * prop.table(tab, margin = 2), 1))
}

# 2.4 Categorical variables: test and effect size
# min_expected checks the chi-square assumption (use Fisher if < 5)
cat_tests <- map_dfr(cat_vars, function(v) {
  tab <- table(df[[v]], df$Group)
  ch  <- chisq.test(tab)
  tibble(Variable     = v,
```

```
        Chi2         = unname(ch$statistic),
        df           = unname(ch$parameter),
        p_value      = ch$p.value,
        Cramers_V    = cramers_v(tab)[[1]],
        min_expected = min(ch$expected))
})

print(cat_tests, width = Inf)
# Note: 8 tests in Step 2 -> Bonferroni threshold 0.05 / 8 = 0.00625


# ============================================================================
# STEP 3 — Churn rate per category with Wilson confidence intervals
# Difference from the reference (first) level in percentage points,
# with a 95% CI from prop.test. Numeric variables are banded for display only.
# ============================================================================
library(binom)

# 3.1 Display bands for numeric variables
df <- df %>%
  mutate(
    TicketsBand  = cut(SupportTickets, breaks = c(-Inf, 0, 2, 5, Inf),
                       labels = c("0", "1-2", "3-5", "6+")),
    SessionsBand = cut(AvgMonthlySessions, breaks = c(-Inf, 14, 29, 44, Inf),
                       labels = c("2-14", "15-29", "30-44", "45-59")),
    DiscountBand = factor(DiscountRate, labels = c("0%", "5%", "10%", "15%", "25%"))
  )

# 3.2 Function: churn rate + Wilson CI + difference from reference
rate_table <- function(var) {
  t <- df %>%
    group_by(Level = .data[[var]]) %>%
    summarise(n = n(), churned = sum(Churned), .groups = "drop") %>%
    mutate(Variable = var, Level = as.character(Level), rate = churned / n)

  ci <- binom.confint(t$churned, t$n, methods = "wilson")
  t$lower <- ci$lower
  t$upper <- ci$upper

  t$diff_pp <- NA; t$diff_low <- NA; t$diff_high <- NA
  for (i in seq_len(nrow(t))[-1]) {
    pt <- prop.test(c(t$churned[i], t$churned[1]), c(t$n[i], t$n[1]), correct = FALSE)
    t$diff_pp[i]   <- 100 * (t$rate[i] - t$rate[1])
    t$diff_low[i]  <- 100 * pt$conf.int[1]
    t$diff_high[i] <- 100 * pt$conf.int[2]
  }
  t
}

# 3.3 Apply to all factors
vars3 <- c("AttendedOnboarding", "Plan", "BillingCycle", "PaymentMethod",
           "TicketsBand", "SessionsBand", "DiscountBand")

results3 <- map_dfr(vars3, rate_table) %>%
  select(Variable, Level, n, churned, rate, lower, upper, diff_pp, diff_low, diff_high) %>%
  mutate(across(c(rate, lower, upper), ~ round(100 * .x, 1)),
         across(c(diff_pp, diff_low, diff_high), ~ round(.x, 1)))

print(results3, n = Inf, width = Inf)
```

```r
# ============================================================================
# STEP 4 — Multivariable logistic regression
# Step 3 showed a U-shaped relationship for sessions and a rise at high ticket
# counts, so a linear model is compared with a flexible model where sessions
# and tickets enter as natural splines (3 df). MonthlyPrice is excluded
# (one fixed price per plan -> perfectly collinear with Plan).
# ============================================================================
library(splines); library(broom); library(marginaleffects); library(car)

# Discount in units of 5 percentage points, so its coefficient is meaningful
df <- df %>% mutate(Discount5 = DiscountRate * 20)

# 4.1 Linear model
m_linear <- glm(Churned ~ Plan + BillingCycle + PaymentMethod + AttendedOnboarding +
                  Discount5 + SupportTickets + AvgMonthlySessions,
                data = df, family = binomial)

# 4.2 Flexible model (splines) — 13 parameters, ~17 churn events per parameter
m_spline <- glm(Churned ~ Plan + BillingCycle + PaymentMethod + AttendedOnboarding +
                  Discount5 + ns(SupportTickets, df = 3) + ns(AvgMonthlySessions, df = 3),
                data = df, family = binomial)

# Is the curvature real? (LRT and AIC) -> spline model chosen
anova(m_linear, m_spline, test = "LRT")
AIC(m_linear, m_spline)

# 4.3 Importance of each factor: drop-one likelihood ratio test
# Ranking uses the AIC increase on deletion (fair across terms with different df)
drop1(m_spline, test = "LRT")

# 4.4 Odds ratios with 95% CI (spline terms excluded: not interpretable alone)
tidy(m_spline, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(!str_detect(term, "ns\\(")) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  print(width = Inf)

# 4.5 Average marginal effects (AME) in percentage points
# Pre-specified contrasts for the curved variables:
#   sessions 30 vs 10 (near overall mean vs low use); tickets 8 vs 2 (2 = median)
ame_cat <- avg_comparisons(m_spline,
             variables = c("AttendedOnboarding", "Plan", "BillingCycle",
                           "PaymentMethod", "Discount5"))

ame_num <- avg_comparisons(m_spline,
             variables = list(AvgMonthlySessions = c(10, 30),
                              SupportTickets     = c(2, 8)))

bind_rows(as.data.frame(ame_cat), as.data.frame(ame_num)) %>%
  transmute(term, contrast,
            AME_pp = round(100 * estimate, 1),
            low    = round(100 * conf.low, 1),
            high   = round(100 * conf.high, 1),
            p.value = signif(p.value, 3)) %>%
  print()

# 4.6 Multicollinearity (GVIF^(1/(2*Df)) near 1 = no problem)
vif(m_spline)
```

```r
# 4.7 Shape of the relationship after adjustment
# Note: other variables are held at typical values, so read the SHAPE, not the level
plot_predictions(m_spline, condition = "AvgMonthlySessions")
plot_predictions(m_spline, condition = "SupportTickets")

# 4.8 Post-hoc check of the upturn at heavy use (55 vs 28 sessions)
# Exploratory: 28 was chosen after seeing the curve (its minimum)
avg_comparisons(m_spline, variables = list(AvgMonthlySessions = c(28, 55)))


# ============================================================================
# STEP 5 — Model evaluation (stratified 5-fold cross-validation)
# Each subscriber is scored by a model that did not see them.
# Accuracy alone is misleading with 9.3% churn, so PR-AUC, recall, precision,
# Brier score and calibration are reported alongside ROC-AUC.
# ============================================================================
library(pROC); library(PRROC)

# 5.1 Stratified folds (same churn rate in each fold)
set.seed(2026)
k <- 5
df <- df %>%
  group_by(Churned) %>%
  mutate(fold = sample(rep(1:k, length.out = n()))) %>%
  ungroup()

f_spline <- formula(m_spline)
df$pred_cv <- NA_real_

for (i in 1:k) {
  train    <- df %>% filter(fold != i)
  test_idx <- which(df$fold == i)
  m <- glm(f_spline, data = train, family = binomial)
  df$pred_cv[test_idx] <- predict(m, newdata = df[test_idx, ], type = "response")
}

# 5.2 Discrimination
roc_obj <- roc(df$Churned, df$pred_cv, quiet = TRUE)
cat("ROC-AUC:", round(auc(roc_obj), 3), "\n")
print(ci.auc(roc_obj))

pr <- pr.curve(scores.class0 = df$pred_cv[df$Churned == 1],
               scores.class1 = df$pred_cv[df$Churned == 0], curve = TRUE)
cat("PR-AUC:", round(pr$auc.integral, 3),
    " | Baseline (churn rate):", round(mean(df$Churned), 3), "\n")

# 5.3 Calibration
brier      <- mean((df$pred_cv - df$Churned)^2)
brier_null <- mean((mean(df$Churned) - df$Churned)^2)
cat("Brier:", round(brier, 4), " | Brier of no-variable model:", round(brier_null, 4), "\n")

df %>%
  mutate(decile = ntile(pred_cv, 10)) %>%
  group_by(decile) %>%
  summarise(n = n(),
            predicted = round(100 * mean(pred_cv), 1),
            observed  = round(100 * mean(Churned), 1)) %>%
  print(n = Inf)
```

```r
# 5.4 Accuracy vs recall and precision at different thresholds
thr_metrics <- function(thr, label) {
  flag <- as.integer(df$pred_cv >= thr)
  tp <- sum(flag == 1 & df$Churned == 1)
  fp <- sum(flag == 1 & df$Churned == 0)
  fn <- sum(flag == 0 & df$Churned == 1)
  tibble(rule      = label,
         threshold = round(thr, 3),
         flagged   = sum(flag),
         recall    = round(tp / (tp + fn), 3),
         precision = round(tp / (tp + fp), 3),
         accuracy  = round(mean(flag == df$Churned), 3))
}

bind_rows(
  tibble(rule = "No one churns (baseline)", threshold = NA, flagged = 0,
         recall = 0, precision = NA, accuracy = round(1 - mean(df$Churned), 3)),
  thr_metrics(0.5, "Default threshold 0.5"),
  thr_metrics(mean(df$Churned), "Threshold = churn rate"),
  thr_metrics(quantile(df$pred_cv, 0.8), "Top 20% risk")
) %>% print(width = Inf)

# 5.5 Precision-Recall curve (dashed line = baseline)
plot(pr, main = "Precision-Recall (CV)")
abline(h = mean(df$Churned), lty = 2)


# ============================================================================
# STEP 6 — Testing a proposed mechanism
# Hypothesis: not attending onboarding leads to more support tickets.
# Result: not supported (means 2.78 vs 2.80, p = 0.93) -> the two factors
# are associated with churn independently.
# ============================================================================
df %>%
  group_by(AttendedOnboarding) %>%
  summarise(n = n(),
            mean_tickets   = round(mean(SupportTickets), 2),
            median_tickets = median(SupportTickets),
            pct_6plus      = round(100 * mean(SupportTickets >= 6), 1))

wilcox.test(SupportTickets ~ AttendedOnboarding, data = df, exact = FALSE)


# ============================================================================
# Session info (for reproducibility)
# ============================================================================
sessionInfo()
```